



Research paper

FTO-mediated m6A demethylation inhibits bladder cancer progression via decreasing EMG1 and reducing ribosome biosynthesis

Kun Yao, Long Wang, Jinrong Wang, Jianye Liu, Chao Li, Weibin Hou, Bingzhi Wang, Wei Xiong, Jing Tan*

Department of Urology, The Third Xiangya Hospital, Central South University, Changsha, Hunan, 410013, China

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ABSTRACT

Bladder cancer (BLCA) is one of the most common urological malignancies, entailing significant morbidity and mortality rates. A comprehensive understanding of the mechanisms driving bladder cancer initiation and development is required to devise improved treatment regimens. N6-methyladenosine (m6A) is the most prevalent internal RNA modification reported to regulate cancer metastasis. Here, the function of fat mass and obesity-associated protein (FTO), an m6A demethylase, was studied in bladder cancer by overexpressing or knocking down FTO, and the underlying mechanism of FTO in BLCA was explored using machine learning analysis, in vitro, and in vivo experiments. The FTO expression has been proven to be downregulated within bladder cancer, and it could exert a tumor-suppressive effect. Moreover, Gain- and loss-of-function experiments showed that FTO downregulation enhanced the proliferation, migration, and invasion of bladder cancer cells. Mechanistic studies revealed that FTO decreased EMG1 expression by demethylating EMG1 and reducing ribosome biosynthesis, thereby promoting bladder cancer cell proliferation, migration, and invasion and repressing tumor growth in vivo. This study has demonstrated the anti-tumor effect of FTO on bladder cancer development, making it a promising therapeutic target.

1. Introduction

Bladder cancer (BLCA) is the most prevalent malignancy of the urinary system with increasing morbidity and mortality rates [1,2]. Although BLCA patients' prognosis was improved due to the application of radiotherapy, chemotherapy, and targeted therapy, the tumor has a high recurrence and metastasis rate, and the prognosis remains impaired [2,3]. Therefore, understanding the key factors and potential mechanisms associated with bladder cancer progression is both urgent and essential.

N6-Methyladenosine (m6A) is the most abundant modification of eukaryotic mRNA and involves almost all aspects of RNA metabolism [4–9]. m6A-mediated epigenetic modification plays a critical role in the occurrence and development of multiple malignancies, including bladder cancer [10–12]. For example, M6A-mediated upregulation of lncRNA BLACAT3 enhances angiogenesis, epithelial-mesenchymal transition (EMT), and metastasis in bladder cancer [12]. METTL3 promotes the proliferation of bladder cancer by promoting pri-miR221/222 maturation in an m6A-dependent manner [13]. m6A-induced lncDBET

promotes the malignant progression of bladder cancer [14]. Fat mass and obesity-associated protein (FTO), an eraser of m6A, exerts a complex effect on bladder cancer [15–17]. FTO downregulation enhances the proliferation and migratory ability of bladder cancer cells and protects them against cisplatin-induced cytotoxicity [17]. Other researchers reported that FTO enhances bladder cancer cell proliferation through the FTO/miR-576/CDK6 pathway in a m6A-dependent manner [18]. Therefore, further confirming the role and underlying mechanism of FTO in the initiation and progression of bladder cancer is important.

Ribosome biogenesis is a highly complex process that takes place in the nucleolus and culminates in the cytoplasm, involving the assembly of ribosomal complexes. This process is tightly related to protein formation, cell proliferation, differentiation, and apoptosis [19,20]. This complex process encompasses rRNA transcription, splicing, and modification, followed by ribosome assembly and pre-ribosome export. It involves the coordinated activity of 3 RNA polymerases, around 80 ribosomal proteins, and about 200 non-ribosomal transcription factors. Since the excessive proliferation and other phenotypes of cancer cells require a greater amount of energy supply than normal cells, overactive

* Corresponding author.

E-mail address: tanjing@csu.edu.cn (J. Tan).

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ribosome biogenesis drives tumor cell growth and proliferation, becoming a core factor in cancer occurrence and metastasis [21–23]. Ribosome biogenesis is regulated by various factors, including mammalian targets of rapamycin (mTOR), myelocytomatosis oncogene (Myc), noncoding RNA (ncRNA), and m6A, among which m6A is a widely studied one, especially in cancer [20,24]. A recent study confirmed that an increase in m6A levels upregulates ribosome biogenesis in environmental carcinogenesis, and FTO can regulate this process [25]. However, the regulation between m6A and ribosome biogenesis in bladder cancer remains unclear.

Herein, the function of FTO in bladder cancer was determined by overexpressing or knocking down FTO and exploring whether FTO exerted its functions in a ribosome biogenesis-related manner in BLCA using bioinformatics analysis, as well as in vitro and in vivo experiments. Our findings might provide a viable therapeutic target for bladder cancer.

2. Materials and methods

2.1. Public data acquisition and preprocessing

The gene expression profiles and corresponding clinical data of bladder cancer were collected from public databases, including Gene Expression Omnibus (GEO, <https://www.ncbi.nlm.nih.gov/geo/>) and The Cancer Genome Atlas (TCGA, <https://portal.gdc.cancer.gov/>). The genes related to ribosome biosynthesis were obtained from the GSEA database (<https://www.gsea-msigdb.org/gsea/msigdb/>). The m6A-negative regulatory target of FTO was obtained from the RM2Target database (<http://rm2target.canceromics.org/#/home>), intersected with the ribosome-generating gene, and subjected to differential analysis using absolute Log_2FC values ≥ 0.585 and $\text{FDR} < 0.05$. The differential expression and survival status of FTO in the TCGA-BLCA cohort were subsequently analyzed using the GSCA database (<https://guolab.wch.scu.cn/GSCA/#/>). GSE188715 ($n = 70$) was also used for the analysis of FTO and EMG1 expression. The raw count data were normalized using the voom function in the R package ‘limma’ (Version: 3.62.1). The Venn Diagrams were plotted using VENNEY2.1.0 (<https://bioinfo.gp.cnb.csic.es/tools/venny/>) and Hello jvenn! (<https://jvenn.toulouse.inra.fr/app/example.html>).

2.2. Functional enrichment analysis

The ‘enrichplot’ R package was used to calculate KEGG and GO (Gene Ontology) enrichment scores for core genes in circadian rhythms and to plot relevant pathway diagrams. The Gene Set Enrichment Analysis (GSEA) method and the ‘enrichplot’ R package were used to calculate the functional enrichment scores of GO subtypes. The infiltration levels of 22 immune cell markers in each TCGA-BLCA cancerous specimen were quantified with merged expression data using the CIBERSORT algorithm.

2.3. Machine learning analysis

Five machine-learning algorithms were used to identify bladder cancer-related genes: support vector machine model SVM [26], random forest model RF [27], logistic regression model Logit [28], K-nearest neighbor algorithm KNN [28], and decision tree model DTS [29]. ROC analysis was performed on five machine learning models, and algorithm genes with AUC values greater than 0.9 were selected. The intersection was taken to obtain key genes.

2.4. Clinical sample collection

Twelve paired of bladder cancer (transitional cell carcinoma) and non-cancerous clinical tissue (2 cm from the tumor edge) samples were collected from patients receiving surgical resection in The Third Xiangya

Hospital of Central South University. All procedures were approved by the Institutional Review Board of The Third Xiangya Hospital of Central South University (No. 2021-S310). All patients signed the informed consent forms. Tissues were immediately stored at -80°C .

2.5. Cell culture

The human bladder cancer cell lines HT1197, HT1376, SW780, T24, and 6537 were collected from the American Type Culture Collection (ATCC, Manassas, VA, USA). The immortalized human ureteral epithelial cell line SV-HUC1 was also collected from ATCC. Cells were cultivated within Dulbecco's modified Eagle's (DMEM, 30030, Thermo Fisher Scientific, Waltham, MA, USA) RPMI-1640 (11875093, Thermo Fisher Scientific) media or DMEM/F12 (11330032, Thermo Fisher Scientific) containing 10% FBS (A5670801, Thermo Fisher Scientific) and penicillin-streptomycin antibiotics (15070063, Thermo Fisher Scientific) and kept at 37°C in a 5% CO_2 atmosphere.

2.6. Cell transfection

FTO overexpression plasmids and plasmids containing short hairpin RNA targeted FTO (sh-FTO) were designed and constructed by GenePharma Corporation (Shanghai, China). OriTrans®PEI-DNA (Ori-bio, Changsha, China) was used to transiently transfect the FTO-overexpressing plasmid or shRNA plasmid into HT1197 and HT1376 cell lines. Forty-eight hours later, Western blot and qRT-PCR analyses were performed to verify overexpression or specific silencing effects. All sequences are listed in Supplementary Table 1.

2.7. RNA extraction and qPCR analysis

TRIzol reagent (15596018CN, Thermo Fisher Scientific) was used to extract total RNA from bladder cancer tissue samples or cell lines. The Advantage RT-for-PCR Kit (4368814, Thermo Fisher Scientific) was used to obtain cDNA, and SYBR Premix EX Taq (4309155, Thermo Fisher Scientific) was used to perform quantitative real-time PCR as directed by the manufacturer. The $2^{-\Delta\Delta\text{Ct}}$ method was used for data analysis. Supplementary Table 2 lists all sequences. For mRNA stability analysis, cells were incubated with actinomycin D (10 $\mu\text{g/mL}$, Sigma-Aldrich) for 2, 4, and 6 h, and the mRNA levels of EMG1 were tested by RT-qPCR.

2.8. Western blot

Total proteins were extracted from the cell lines and tissues. Following electrophoresis by SDS-polyacrylamide gel electrophoresis, the separated proteins were electroblotted from the gel onto PVDF membranes (IPVH00010, Millipore, Billerica, MA, USA). Membranes were blocked for 1 h at room temperature (RT) with 5% skim milk, then incubated overnight at 4°C with primary antibodies, and incubated for 1 h at RT with the secondary antibody. The chemiluminescence and gel imager (ChemiDoc XRS, Bio-Rad, Hercules, CA, USA) was used to detect proteins. Primary antibodies used in this study: anti-FTO (1:1000, ab280081, Abcam, Cambridge, UK), anti-EMG1 (1:1000, 11965-1-AP, Proteintech, Wuhan, China), anti-MMP-9 (1:500, 10375-2-AP, Proteintech), anti-N-cadherin (1:500, AF4039, Affinity, Changzhou, China), anti-GAPDH (1:3000, AF7021, Affinity).

2.9. CCK8 assay

The transfected cells were plated in 96-well plates (2000 cells/well) and incubated at 37°C in 5% CO_2 for 0, 24, 48, and 72 h. For each well, the absorbance was assessed at a wavelength of 450 nm at the indicated times.

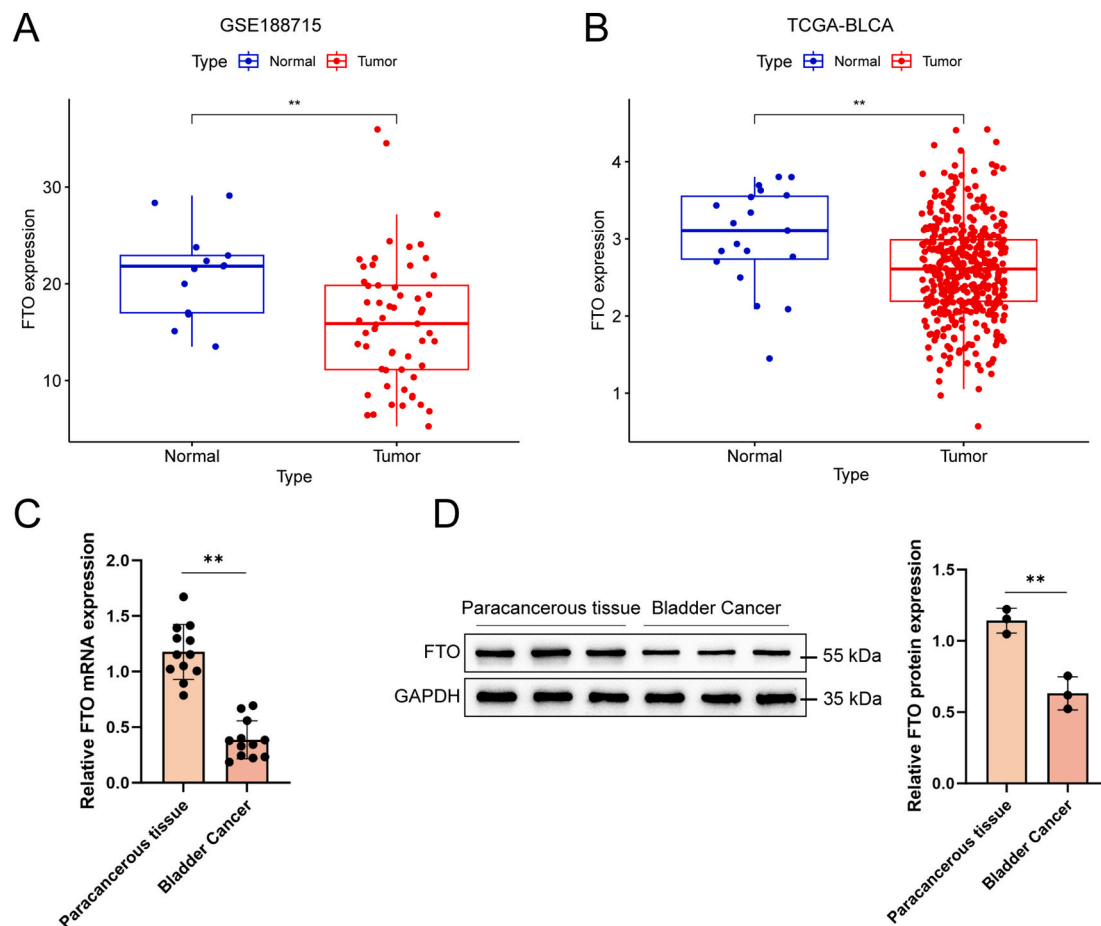


Fig. 1. Differential expression of FTO in bladder cancer. (A) The boxplot of FTO differential expression in the GSE188715 dataset. (B) The boxplot of FTO differential expression in the TCGA-BLCA dataset. (C) The mRNA expression of FTO in bladder cancer and paracancerous tissue was detected by qRT-PCR. $n = 12$ (D) The protein expression of FTO in bladder cancer and paired paracancerous tissue was detected by western blot. $N = 3$. Paired t -test. Data are presented as mean $\pm$ SD, $^{**}P < 0.01$.

2.10. Colony formation assays

The transfected bladder cancer cells were cultured for 10 days on 6-well plates (600 cells per well). Cell colonies were fixed for 20 min with 4% paraformaldehyde and stained for 30 min at RT with 0.1% crystal violet.

2.11. Wound healing assay

The transfected bladder cancer cells were seeded into 6-well plates until reaching 90%–100% confluence. A sterile 200 μ L pipette tip was used to scratch cells, which were photographed at the indicated time. The scratch area was measured using ImageJ.

2.12. Transwell-invasion assay

A 24-well Transwell plate with an 8 μ m pore-size membrane pre-coated with Matrigel (354234, Corning, New York, USA) was used to assess cell invasion. The transfected cell lines (3×10^4 cells) were planted onto the top chamber of the 24-well Transwell plates with serum-free media, and the bottom chambers were supplemented with 800 μ L of media containing 20% FBS. 24 h later, a cotton swab was utilized to scrape the cells at the top of the insert. The cells at the bottom were subsequently fixed with paraformaldehyde and stained with 0.1% crystal violet. A microscope was used to count invaded cells.

2.13. Ethynyl uridine (EU) assay

The Cell-Light EU RNA Imaging Kit (C10316-1, RiboBio, Guangzhou, China) was used to measure nascent RNA synthesis. Cells were subjected to 3-h labeling using 1 mM 5-ethynyl uridine. The EU-labeled RNA was subsequently determined as per the instructions. A fluorescence microscope (Olympus) was applied to observe fluorescent signals to quantify new RNA synthesis.

2.14. Cellular m6A level measurement

Total RNA from bladder cancer cells was isolated as described above. The total m6A methylation amount was assessed using the EpiQuik m6A RNA Methylation Quantification Kit (P-9005; Epigentek Group Inc., NY, USA) as directed by the manufacturer. In this procedure, RNA samples (200 ng) along with m6A standards were immobilized onto the assay plate. Sequential incubations were performed using specific antibodies for m6A capture and detection. The OD value of each well was determined at 450 nm. The methylation levels were calculated by the standard curve.

2.15. MeRIP-qPCR

MeRIP-qPCR was carried out according to the riboMeRip m6A Transcript Profiling Kit (C11051-1, RiboBio, Guangzhou, China). RNA Fragmentation Reagents (AM8740, Invitrogen, Carlsbad, California, USA) were used to fragment purified mRNA into approximately 100-nucleotide fragments, followed by incubation at 70 $^{\circ}$ C. For RNA

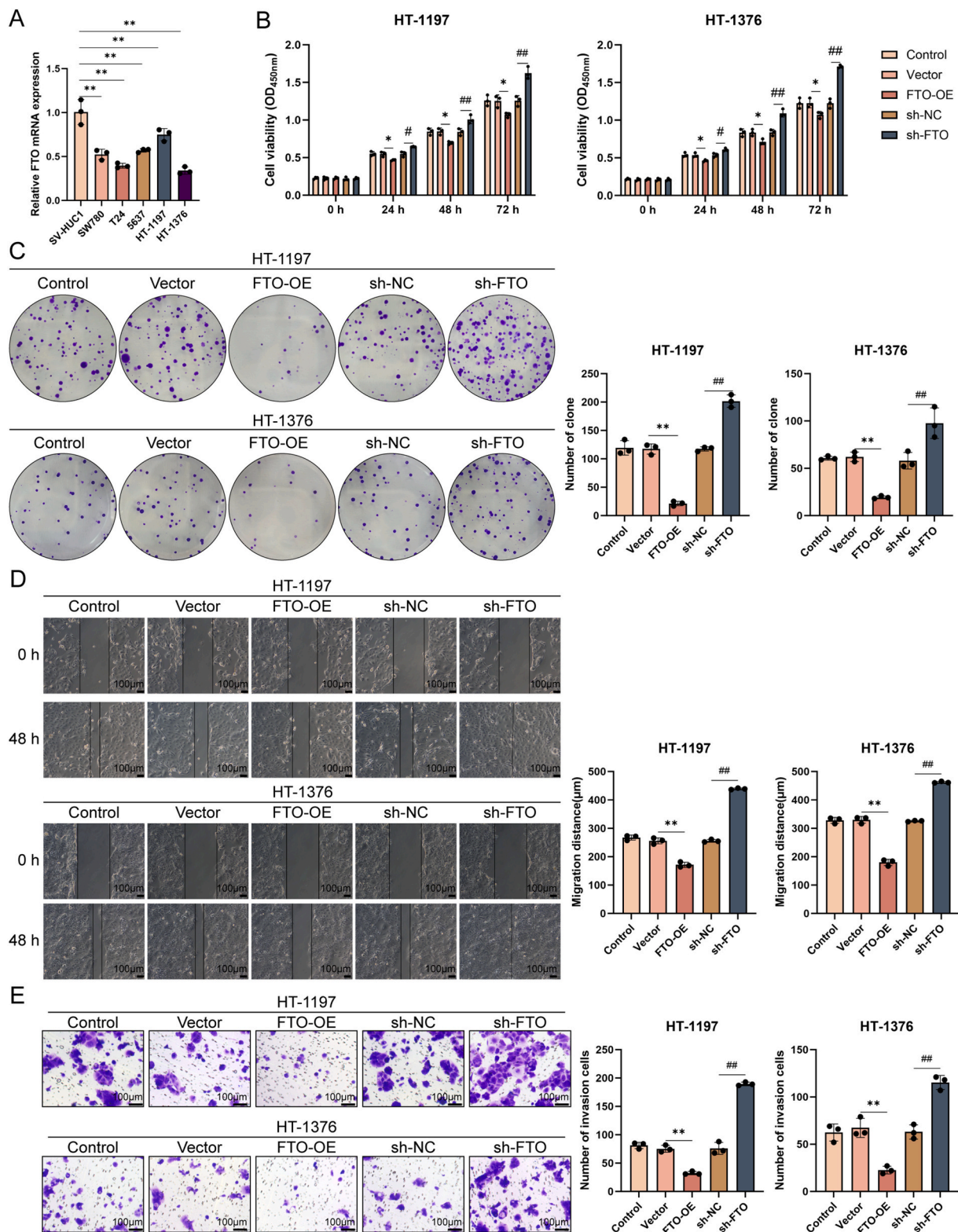


Fig. 2. FTO inhibited proliferation, migration, and invasion of bladder cancer cells. (A) The mRNA levels of FTO in the immortalized human ureteral epithelial cell line SV-HUC1 and human bladder cancer cell lines SW780, T24, 5637, HT-1197 and HT-1376 were determined by qRT-PCR. (B–D) The assay was performed in bladder cancer cells (HT-1376 and HT-1197) with FTO overexpression or knockdown. Cell viability was measured by CCK8 assay (B). The proliferative capability was detected by clone formation assay (C). Cell migration ability was detected by wound healing assay (D). Cell invasion ability was detected by transwell-invasion assay (E). Scale bar 100 μm . Data are presented as mean $\pm$ SD ($N = 3$), One-way ANOVA followed by Tukey's post hoc test. ** $P < 0.01$, compared with the Vector group; ## $P < 0.01$, compared with the sh-NC group.

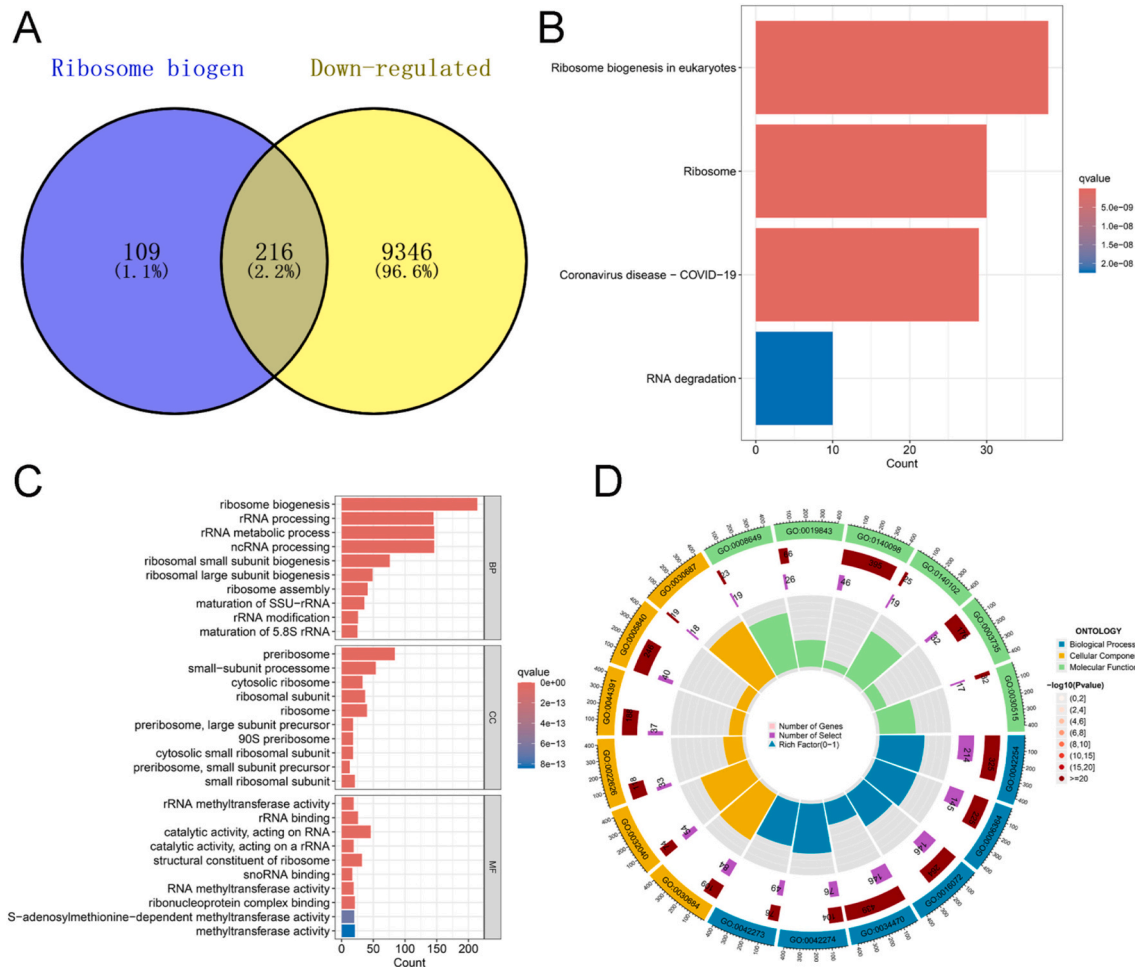


Fig. 3. FTO regulates genes related to ribosome biogenesis. (A) Venn diagram of ribosome biogenesis genes and m6A negative regulatory genes of FTO. (B) KEGG enrichment analysis of key genes involved in ribosome biogenesis. (C) GO enrichment analysis histogram of key genes involved in ribosome biogenesis. (D) GO enrichment analysis circle diagram of key genes involved in ribosome biogenesis.

fragmentation, A/G magnetic beads were used to immunoprecipitate with IgG and m6A antibodies, followed by elution of the beads and purification of the targeted RNA. RT-qPCR was used to analyze targeted RNA with abundant m6A sites.

2.16. Animal models

HT-1197 cells were infected with lentiviruses encoding FTO or EMG1 (Lv-FTO-OE, Lv-EMG1-OE, Ori-bio) (MOI 10) using polybrene for 48 h and subsequently selected with puromycin to yield stable over-expression cell lines. Male athymic nude Balb/c mice (aged 4–6 weeks) were procured from SLAC laboratory animal company (Changsha, China), which were then kept within a specific pathogen-free (SPF) environment. Next, nude mice were injected with 100 μ L (5×10^7 cells/mL) of the cell suspension in the axilla. Approximately 14 days post-injection, once the tumors became palpable, tumor sizes were then measured every 7 days. The formula: $1/2 \times \text{length} \times \text{width}^2$, was used to calculate the tumor volumes. 35 days later, after sacrificing the mice, tumors were excised, measured, weighed, and photographed. The tumor tissues were then fixed and subjected to immunochemical (IHC) staining.

2.17. IHC staining

Tumor tissues were fixed in 10% formalin overnight, then processed for paraffin embedding and sectioning. Then, sections were

deparaffinized, rehydrated, antigen-repaired, and blocked as previously described. Sections were further incubated overnight with anti-FTO, anti-EMG1, and anti-MMP-9 (dilution 1:100). All sections were incubated with HRP-labeled secondary antibodies (dilution 1:100), then incubated in freshly prepared DAB reagent (Beyotime, China), and subsequently counterstained with hematoxylin (Beyotime, China). The sections were observed under a light microscope.

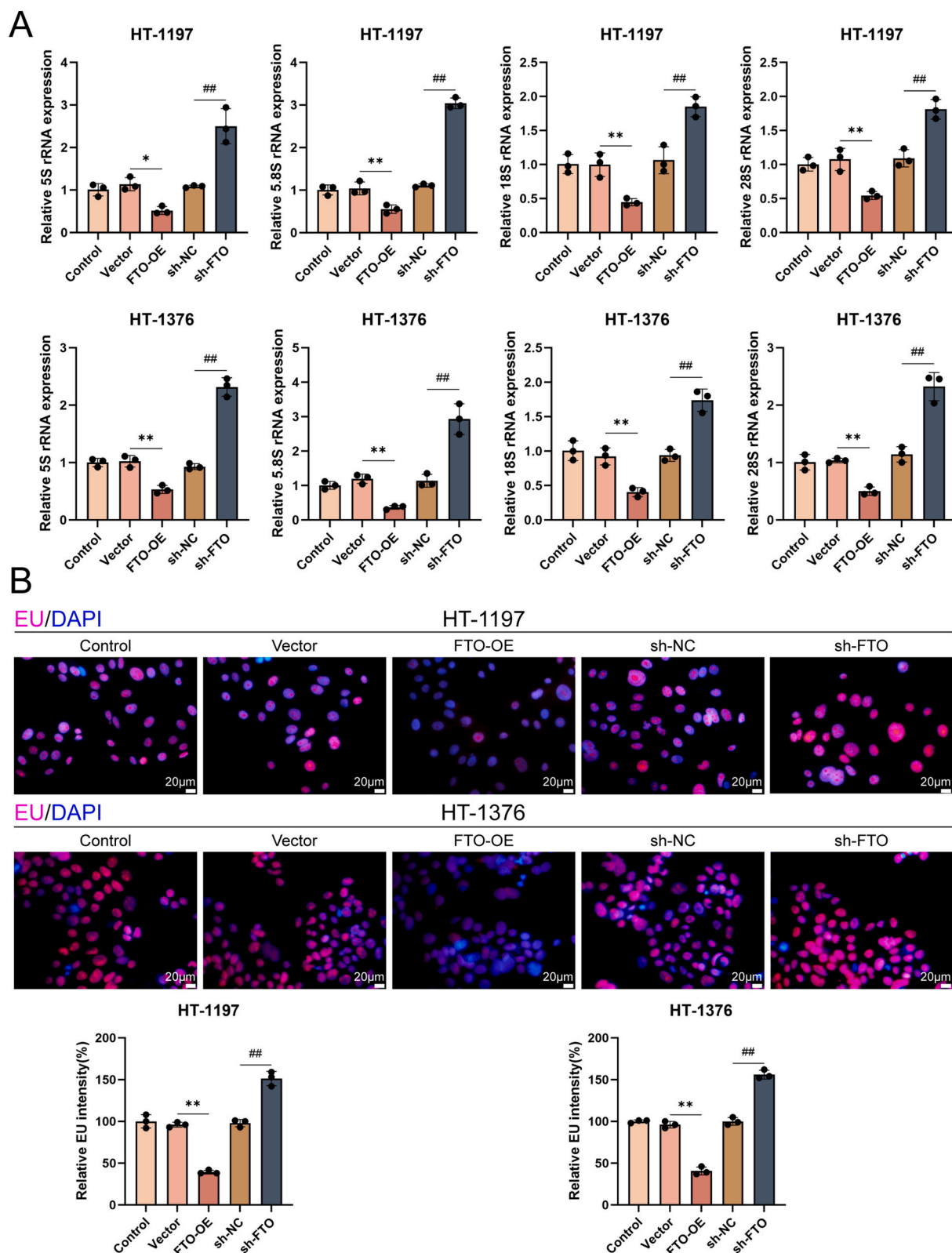
2.18. Statistical analysis

All experimental data were reported as mean $\pm$ standard deviation (SD), and GraphPad Prism 8.0 software was used to conduct one-way analysis of variance (ANOVA) or *t*-tests. The data were evaluated using the Tukey method for multiple comparisons of means. Significance was determined at probability levels of $P < 0.05$.

3. Results

3.1. FTO is downregulated in bladder cancer

To examine FTO expression within the bladder cancer, FTO expression within bladder cancer was assessed based on GSE188715 dataset, and FTO showed to be dramatically decreased within bladder cancer samples in comparison with non-cancerous samples (Fig. 1A). Meanwhile, FTO downregulation was also observed in bladder cancer based on the TCGA-BLCA dataset from the GSCA database (Fig. 1B). Moreover,



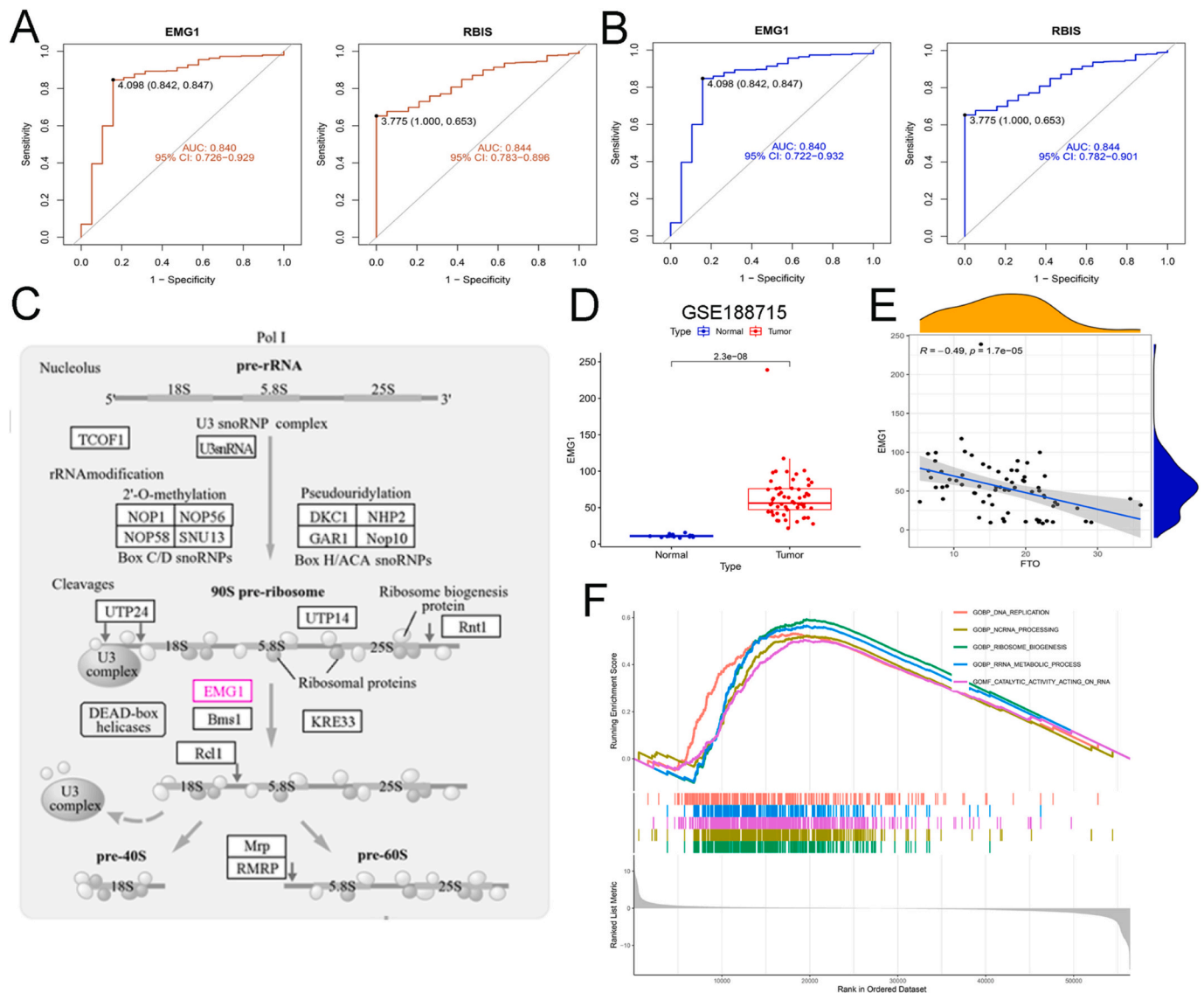


Fig. 5. EMG1 is the core factor of ribosome biogenesis regulated by FTO. (A) ROC curves of EMG1 and RBIS in Logit algorithm. (B) ROC curves of EMG1 and RBIS in SVM algorithm. (C) Ribosomal generation pathway. EMG1 is marked with purple in the picture. (D) Differential expression boxplot of EMG1 in GSE188715 dataset using the limma package. (E) Scatter plot of the Spearman correlation analysis between FTO and EMG1. (F) GSEA analysis chart of EMG1 in TCGA-BLCA dataset.

clinical bladder cancer samples were collected and FTO mRNA level was examined; FTO was dramatically decreased at the mRNA level within bladder cancer tissue samples in comparison with paracancerous tissue samples (Fig. 1C). Consistently, FTO protein levels were also reduced within bladder cancer tissue samples in comparison with paired paracancerous tissue samples (Fig. 1D).

3.2. FTO overexpression inhibits bladder cancer cell proliferation, migration, and invasion

The mRNA expression levels of FTO in bladder cancer cell lines and normal human ureteral epithelial cells were determined, and FTO was downregulated in bladder cancer cell lines compared with SV-HUC1. Moreover, HT-1197 exhibited the highest FTO levels, whereas HT-1376 exhibited the lowest among these bladder cancer cells. Therefore, these two cell lines were used in the following investigation. The efficiency of transfection of FTO-OE or sh-FTO in HT-1197 and HT1376 cells was verified and screened by qRT-PCR (Fig. S1A) and Western blot (Fig. S1B). The results showed that the FTO overexpression cell model was successfully established, and sh-FTO#3 had the best knockdown

efficiency and was chosen for further experiments (Fig. S1). FTO overexpression inhibited bladder cancer cell viability, whereas FTO knockdown resulted in a remarkable elevation within cell viability in comparison with normal controls at 24, 48 and 72 h (Fig. 2B). Additionally, FTO overexpression reduced the cell colonies, and FTO knockdown resulted in an increase in cell colony number, as confirmed by colony formation assays (Fig. 2C). Next, the function of FTO within the migratory and invasive abilities of bladder cancer cells was assessed. Wound-healing and Transwell-invasion assays showed that FTO overexpression was suppressed, whereas FTO knockdown enhanced bladder cancer cell migration and invasion (Fig. 2D and E). Collectively, these data revealed that FTO could inhibit the proliferation, migration, and invasion of bladder cancer cells.

3.3. FTO regulates ribosome biogenesis-related genes

FTO can affect ribosome biogenesis via m6A in environmental carcinogenesis [25]. Therefore, this study explored whether the anti-bladder cancer effect of FTO is related to ribosome biogenesis. A total of 325 ribosome biogenesis-associated genes were downloaded from the

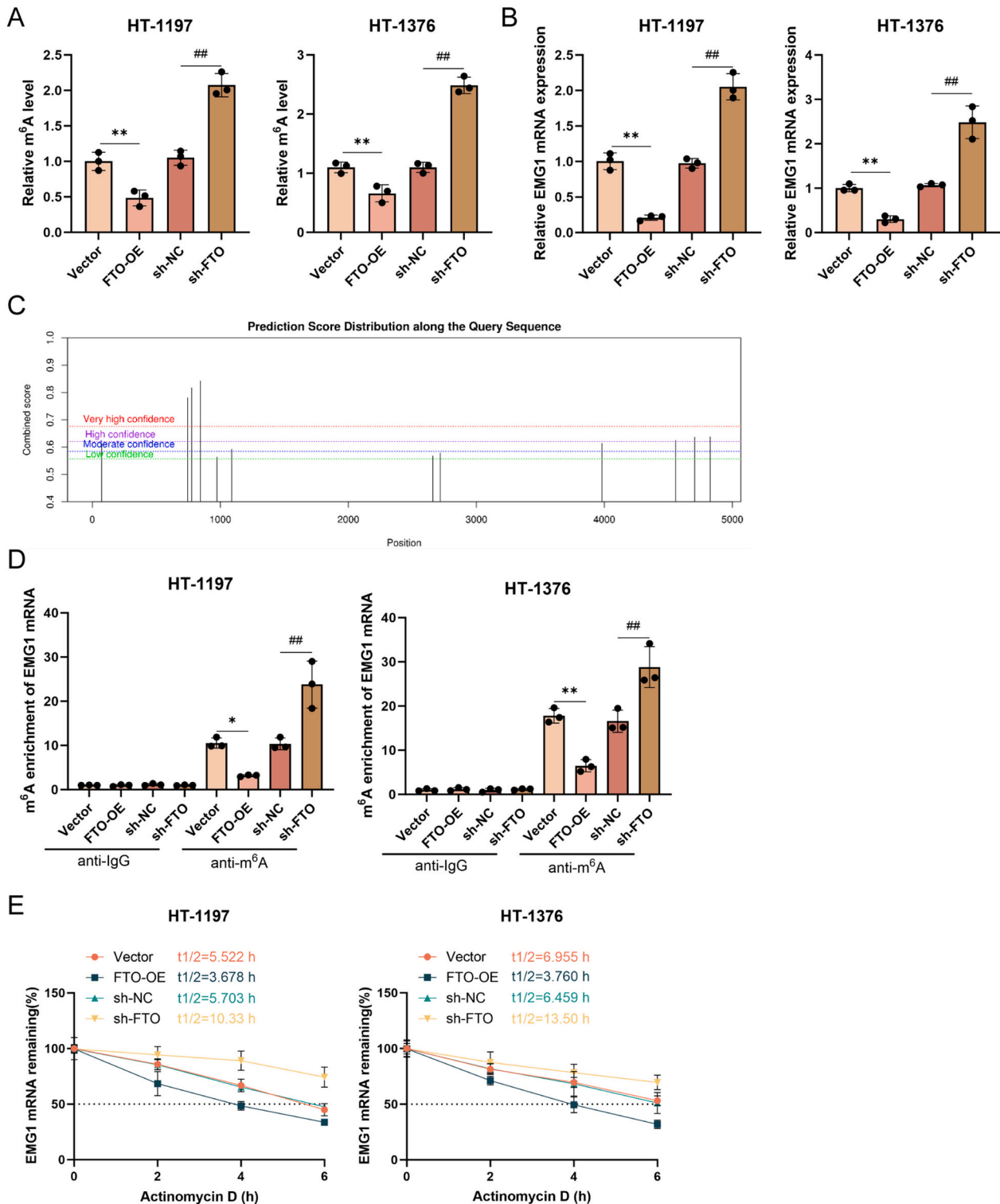


Fig. 6. FTO-mediated demethylation downregulated EMG1. (A) The total m⁶A levels were detected in HT1197 and HT1376 cells using a m⁶A RNA Methylation Quantification Kit. (B) The mRNA expression of EMG1 in bladder cancer cells after FTO overexpression or FTO knockdown was detected by qRT-PCR. (C) Predicting potential M⁶A modification sites of EMG1 using SRAMP database. (D) MeRIP-qPCR detection of m⁶A levels of EMG1. (E) EMG1 levels were measured in HT1197 and HT1376 cells treated with actinomycin D (10 μ g/mL) to inhibit transcription, using qRT-PCR at 0, 2, 4, and 6 h, and calculating decay rate (half-life). Data are presented as mean $\pm$ SD ($N = 3$), One-way ANOVA followed by Tukey's post hoc test. * $P < 0.05$, ** $P < 0.01$, compared with the Vector group; ## $P < 0.01$, compared with the sh-NC group.

GSEA database and intersected with the m⁶A negative regulatory genes of FTO, resulting in 216 key genes associated with ribosome biogenesis (Fig. 3A). KEGG and GO enrichment analysis were performed on 216 key genes involved in ribosome biogenesis. The KEGG results demonstrated

the main enrichment of these genes in ribosome biogenesis and ribosome-related pathways in eukaryotes (Fig. 3B). GO enrichment analysis also revealed the main enrichment of these genes in ribosome-related biological functions (Fig. 3C and D).

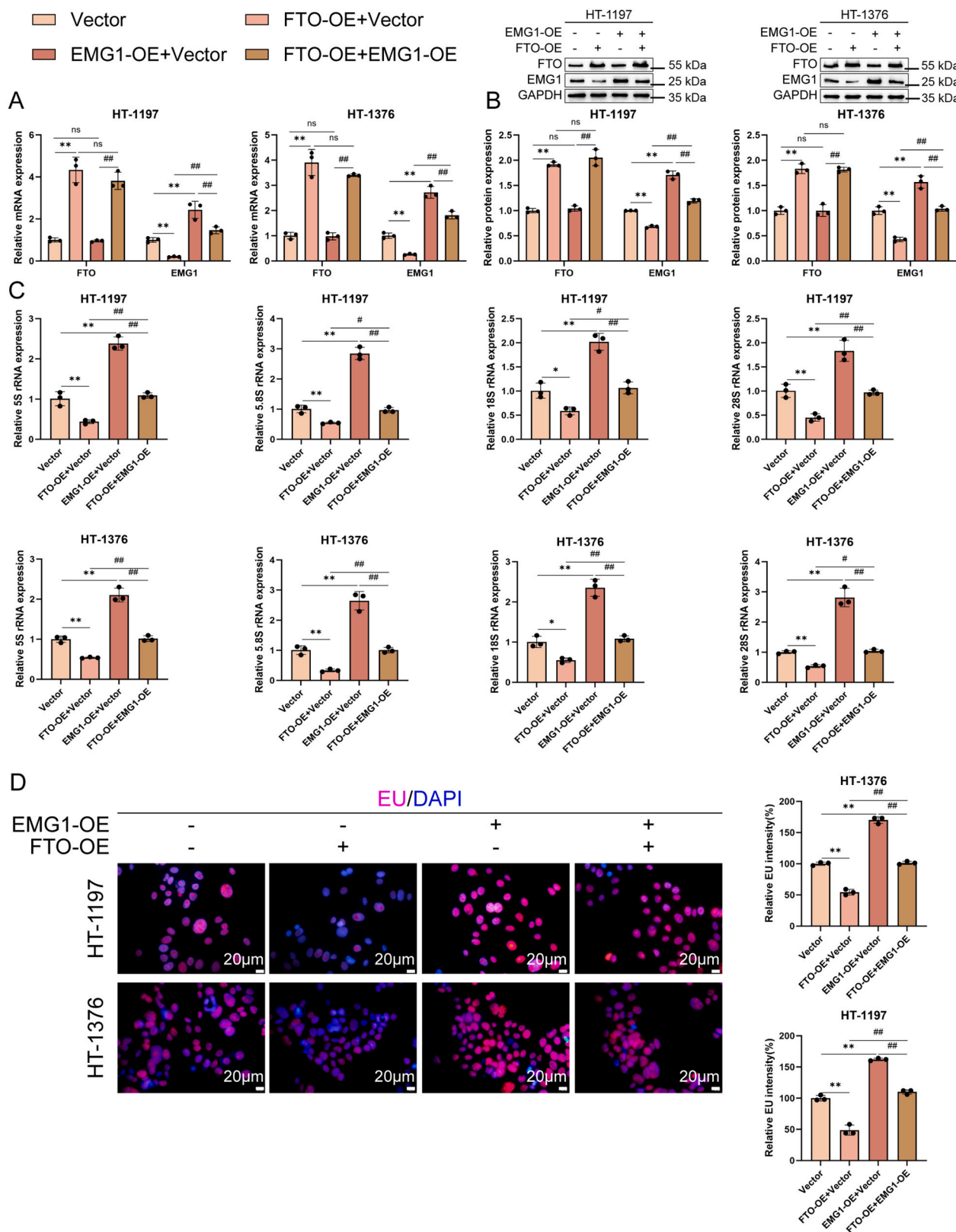


Fig. 7. FTO inhibited cellular ribosome biogenesis by regulating EMG1 levels. The assay was performed in bladder cancer cells (HT-1376 and HT-1197) after overexpression of FTO or/and EMG1. (A) The mRNA expression of FTO and EMG1 was detected by qRT-PCR. (B) The protein expression of FTO and EMG1 was detected by Western blot. (C) rRNA detection of 18S, 5.8S, 28S, and 5S by qRT-PCR. (D) The newly formed RNA was detected by Ethynyluridine (EU) assay. Scale bar 20 μ m. Data are presented as mean $\pm$ SD (N = 3), One-way ANOVA followed by Tukey's post hoc test. * P < 0.05, ** P < 0.01, compared with the Vector group; # P < 0.05, ## P < 0.01, compared with the FTO-OE + EMG1-OE group.

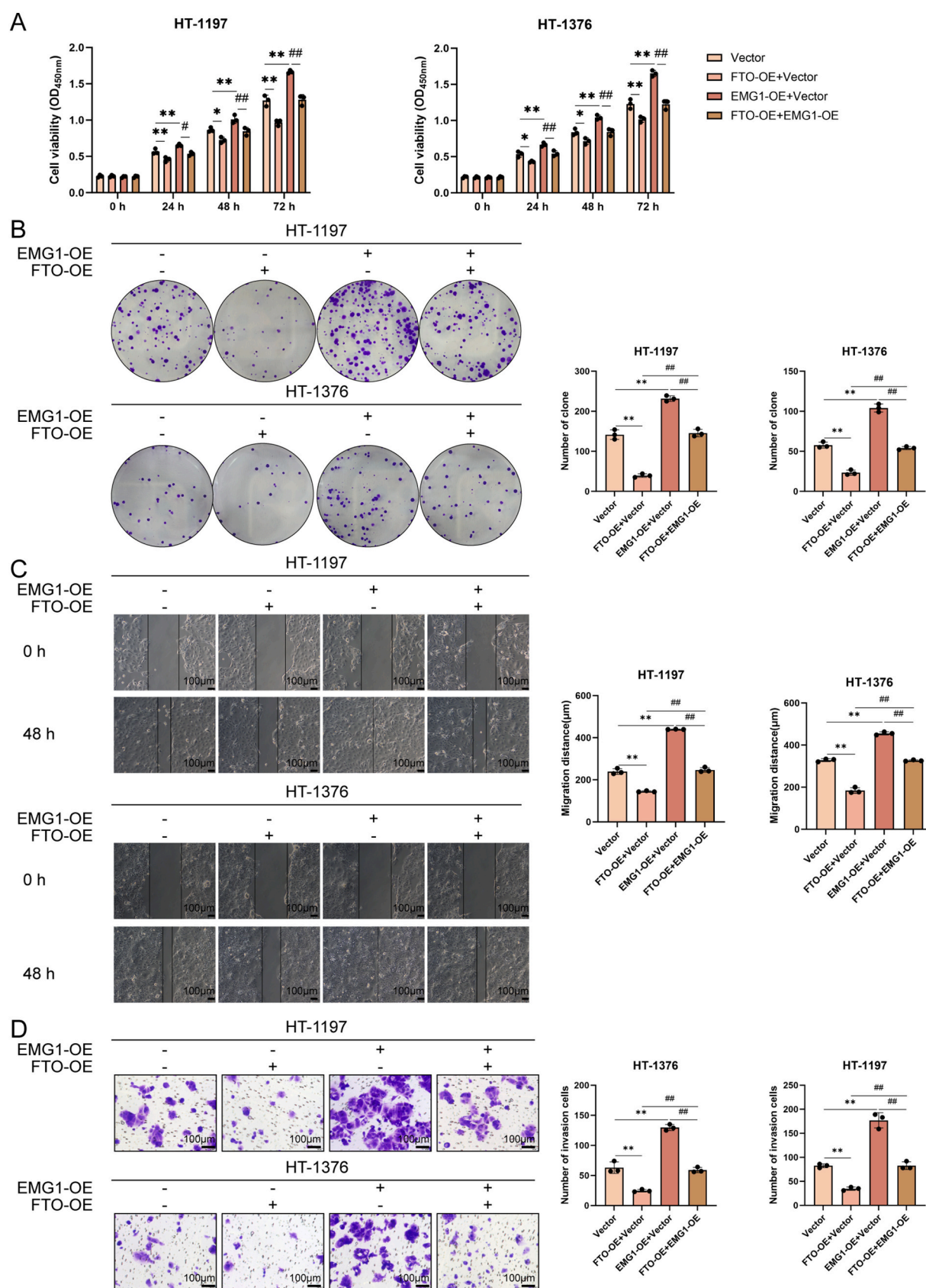


Fig. 8. The proliferation, migration, and invasion of bladder cancer cells were inhibited via FTO-mediated EMG1 downregulation. The assay was performed in bladder cancer cells (HT-1376 and HT-1197) after overexpression of FTO or/and EMG1. (A) Cell viability was measured by CCK8 assay. (B) The proliferative capability was detected by clone formation assay. (C) Cell migration ability was detected by wound healing assay. (D) Cell invasion ability was detected by transwell-invasion assays. Scale bar 100 μm. Data are presented as mean ± SD (N = 3). One-way ANOVA followed by Tukey's post hoc test. * $P < 0.05$, ** $P < 0.01$, compared with the Vector group; # $P < 0.05$, ## $P < 0.01$, compared with the FTO-OE + EMG1-OE group.

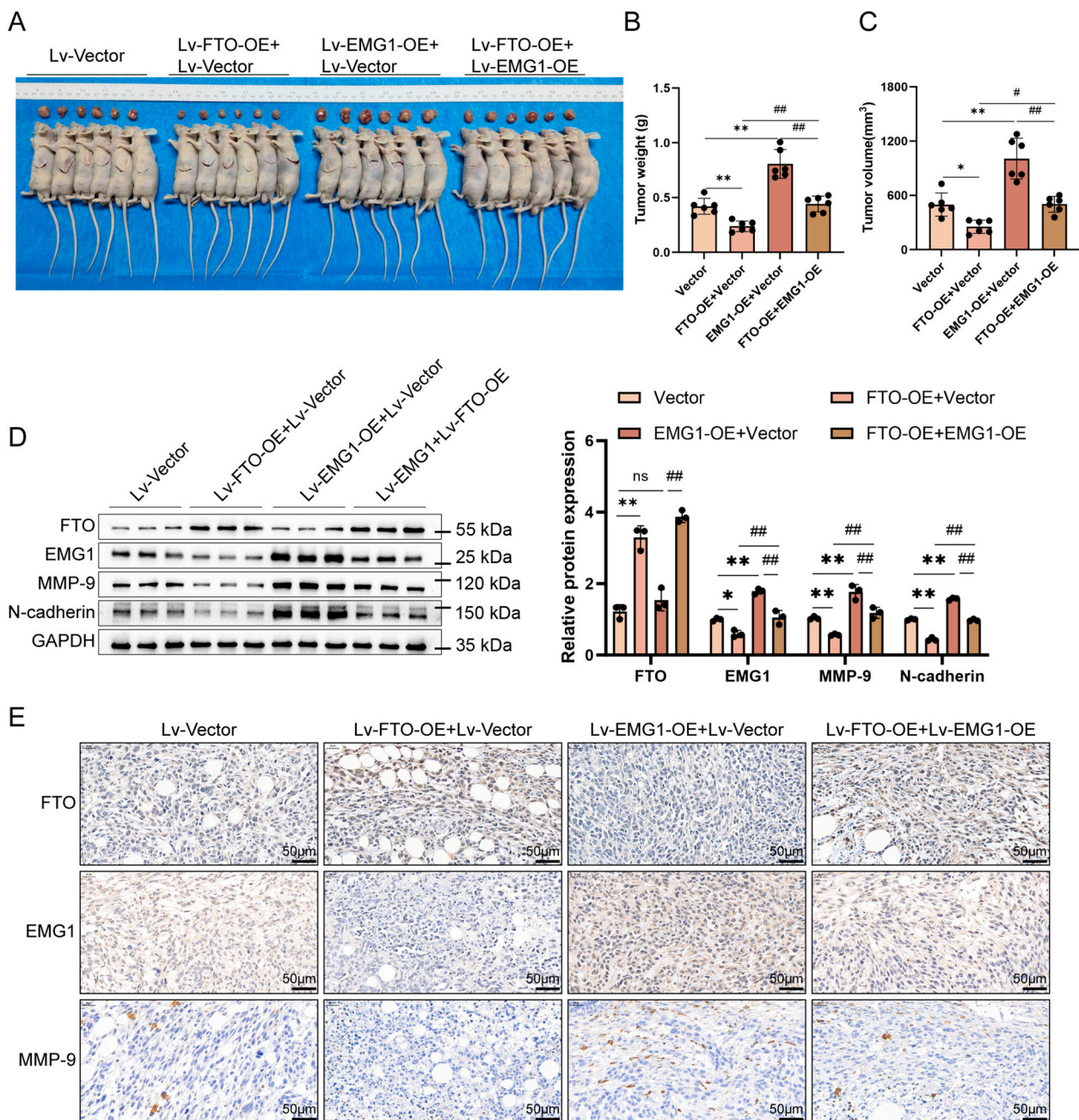


Fig. 9. FTO inhibited tumor growth in bladder cancer in vivo. The control and transfected cells were collected and injected subcutaneously into the left axilla of nude mice. The tumor volume was measured every 7 days from day 14 after injection. Mice were anesthetized and euthanized on day 35, and the tumor tissues were collected for subsequent detection ($n = 6$). (A) The tumor pictures. (B) The tumor weight. (C) The tumor volume. (D) The protein expression of FTO, EMG1, MMP-9, and N-cadherin within the tumor tissues was detected by Western blot. (E) The protein expression of FTO, EMG1 and MMP-9 within the tumor tissues were detected by IHC staining. Scale bar 50 μ m. Data are presented as mean $\pm$ SD ($N = 3$ for D, $N = 6$ for B and C), One-way ANOVA followed by Tukey's post hoc test. $*P < 0.05$, $**P < 0.01$, compared with the Vector group; $\#P < 0.05$, $\#\#P < 0.01$, compared with the FTO-OE + EMG1-OE group.

3.4. FTO overexpression inhibits ribosome biogenesis in bladder cancer cells

To further explore the effect of FTO on ribosome biogenesis, we determined whether FTO overexpression or knockdown influences the rRNA transcript levels. As shown by qRT-PCR analysis, 18 S, 5.8 S, 28 S, and 5 S rRNA levels were suppressed by FTO overexpression, but increased by FTO knockdown in HT1197 and HT1376 cells (Fig. 4A). Moreover, the nascent RNA, particularly rRNA, was labeled using the EU assay. Within HT1197 and HT1376 cell lines, FTO overexpression significantly inhibited the production of nascent rRNAs in the nucleus,

while FTO knockdown promoted nascent rRNA synthesis (Fig. 4B). These results collectively indicate that FTO inhibits rRNA synthesis.

3.5. EMG1 is the core factor of ribosome biogenesis regulated by FTO

To investigate the core factor in ribosome biogenesis regulated by FTO, a differential analysis of key ribosome biogenesis genes was performed on the TCGA-BLCA dataset. Five machine learning algorithms, SVM, RF, Logit, KNN, and DTS, were used to identify bladder cancer disease-related genes in the gene expression profile of the TCGA-BLCA dataset (Fig. S2A). The top 20 genes with the highest scores from each

algorithm were selected as candidate disease-related genes (Fig. S2B, Supplementary Table 3). At the same time, the top 50 differentially expressed genes were displayed as a heat map (Fig. S2F). ROC analysis was performed on the disease-related genes obtained by five machine learning methods. The higher the AUC value, the more accurate the prediction of the model gene for the disease (Fig. S2C). Genes obtained by the Logit and SVM algorithms with AUC values greater than 0.9 were selected, and their intersection was used to identify four key disease genes: EMG1, WDR46, RBIS, and DDX3X (Fig. S2D). Three candidate genes were ultimately identified: EMG1 and RBIS were up-regulated, while DDX3X was down-regulated in bladder cancer (Fig. S2E). Since FTO is downregulated in TCGA-BLCA and the key genes for ribosome biogenesis identified earlier are negatively regulated by FTO, the upregulated genes EMG1 and RIBS were selected for further investigations. In the Logit algorithm, the AUC values of the ROC curves of EMG1 and RBIS are both greater than 0.7, indicating that they are the key factors for bladder cancer (Fig. 5A). Similarly, it is also shown in the ROC results of the SVM algorithm (Fig. 5B). Pathway enrichment analysis on EMG1 and RBIS was subsequently performed. Only EMG1 was enriched in the ribosome biogenesis pathway, while RBIS was not (Fig. 5C). Therefore, EMG1 was chosen as the research object for further study. EMG1 was also upregulated in bladder cancer tissues based on the GSE188715 dataset (Fig. 5D). As shown by Spearman correlation analysis, FTO expression was negatively correlated with EMG1 expression (Fig. 5E). Additionally, GSEA indicated that EMG1 activates biological functions related to ribosome biogenesis based on the TCGA-BLCA dataset (Fig. 5F). These data suggested that EMG1 might serve as a core factor in FTO-mediated regulation of ribosome biogenesis.

3.6. FTO-mediated demethylation downregulated EMG1

To investigate whether FTO affects the expression of EMG1, the total m6A level and EMG1 mRNA levels were measured in HT1197 and HT1376 cells following FTO knockdown/overexpression, indicating that the overexpression of FTO decreased the total m6A level. In contrast, FTO knockdown increased the total m6A level (Fig. 6A). Consistently, FTO overexpression reduced EMG1 mRNA level, while FTO knockdown was elevated (Fig. 6B). Then, as shown in Fig. 6C, the potential m6A modification sites in EMG1 were also predicted using the SRAMP database. Next, EMG1 m6A level within HT1197 and HT1376 cell lines following FTO knockdown or overexpression were assessed using MeRIP-qPCR; FTO overexpression reduced, while FTO knockdown elevated EMG1 m6A level (Fig. 6D). Furthermore, the effect of FTO overexpression or knockdown upon EMG1 mRNA stability was evaluated; FTO overexpression reduced, while FTO knockdown elevated EMG1 mRNA half-life (Fig. 6E). Taken together, these data suggested that FTO downregulates EMG1 mRNA stability through demethylation.

3.7. FTO overexpression inhibits cellular ribosome biogenesis by reducing EMG1 levels

To investigate whether FTO inhibits cellular ribosome biogenesis by reducing EMG1, FTO, and EMG1 were overexpressed in HT1197 and HT1376 cells, and the impact of EMG1 levels on FTO's inhibitory effects on ribosome biogenesis was assessed. Fig. 7A showed that FTO overexpression inhibited EMG1 mRNA level, whereas overexpressed EMG1 could partially reverse the effect of FTO overexpression alone upon EMG1 mRNA. Consistently, FTO overexpression reduced EMG1 protein contents, whereas overexpressing EMG1 could partially reverse the effect of FTO overexpression alone upon EMG1 protein (Fig. 7B). EMG1 overexpression has no significant effect on the mRNA and protein levels of FTO (Fig. 7A and B). Moreover, qRT-PCR results revealed that the 18 S, 5.8 S, 28 S and 5 S rRNA levels were suppressed by FTO overexpression, but partially reversed by overexpressing EMG1 in HT1197 and HT1376 cells (Fig. 7C). Moreover, EU assay showed that FTO overexpression remarkably suppressed the production of nascent rRNAs

within the nucleus while overexpressing EMG1 reversed the inhibitory effect of FTO on nascent rRNA synthesis (Fig. 7D). In summary, FTO inhibits cellular ribosome biogenesis by reducing EMG1 levels.

3.8. The bladder cancer cell proliferation, migration, and invasion were inhibited via FTO-mediated EMG1 downregulation

To further evaluate whether EMG1 downregulation via FTO-mediated demethylation affects FTO functions in bladder cancer cells, FTO and EMG1 were shown to be overexpressed, and the effect of EMG1 on mediating FTO's anti-tumor effects was assessed. As shown in Fig. 8A, FTO overexpression inhibited bladder cancer cell viability, whereas overexpressing EMG1 partially relieved the effect of FTO overexpression alone on cell viability. Additionally, FTO overexpression reduced the cell colonies, whereas overexpressing EMG1 could partially alleviate the effect of FTO overexpression alone upon cell colonies (Fig. 8B). Moreover, as revealed by wound healing and Transwell-invasion assays, FTO overexpression repressed the cell migratory and invasive capabilities, whereas overexpressing EMG1 could partially reverse the effect of FTO overexpression alone upon bladder cancer cell migratory and invasive capabilities (Fig. 8C and D). Taken together, FTO-mediated downregulation of EMG1 can repress bladder cancer cell proliferation, migration, and invasion.

3.9. FTO overexpression represses tumor growth in bladder cancer in vivo

To further study the inhibitory effect of FTO upon bladder cancer in vivo, a xenograft tumor model was established, and FTO overexpression and/or EMG1 overexpression was achieved. FTO overexpression inhibited tumor growth in the nude mice model, whereas overexpressing EMG1 reversed the effect of FTO overexpression alone on tumor growth (Fig. 9A). Consistently, FTO overexpression decreased tumor weight and volume, whereas overexpressing EMG1 increased tumor weight and volume (Fig. 9B and C). Additionally, FTO, EMG1, and EMT-associated protein MMP-9, and N-cadherin contents were also detected within tumor tissue samples through western blot; FTO overexpression inhibited the expression of EMG1, MMP-9, and N-cadherin, whereas overexpressing EMG1 could reverse this effect (Fig. 9D). The IHC staining also exhibited a similar trend. FTO overexpression increased FTO expression and inhibited the expression of EMG1 and MMP-9, whereas overexpressing EMG1 could reverse this effect without affecting FTO expression (Fig. 9E). Collectively, FTO could inhibit tumor growth via reducing EMG1 expression in a nude mice model in vivo.

4. Discussion

FTO, an RNA m6A demethylase, is commonly dysregulated and exerts crucial effects on multiple types of malignancies, including bladder cancer [30–33]. FTO-mediated demethylation contributes to bladder cell proliferation, migration, and drug resistance [15,16,34,35]. Herein, FTO is downregulated in bladder cancer and represses bladder cancer cell proliferation, migration, and invasion in vitro and inhibits tumor growth in vivo. Mechanistically, FTO inhibited bladder cancer cell proliferation, migratory and invasive capabilities, and tumor growth by FTO-mediated m6A demethylation, which decreases EMG1 and reduces ribosome biosynthesis in bladder cancer.

Previous studies have shown that FTO downregulation promotes cell proliferation and migratory capacity and protects cells against cisplatin-induced cytotoxicity in bladder cancer [17]. Moreover, FTO has been reported to inhibit bladder cancer progression by demethylating the m6A modification in CSNK2A2 and ITGA6 mRNAs [15]. In this study, FTO expression in bladder cancer tissues was evaluated using bioinformatics analysis of TCGA-BLCA and GSE datasets, as well as clinical bladder cancer specimens, and FTO was downregulated in bladder cancer tissues compared with adjacent tissues. While several previous reports have shown contradictory results, with FTO being upregulated in

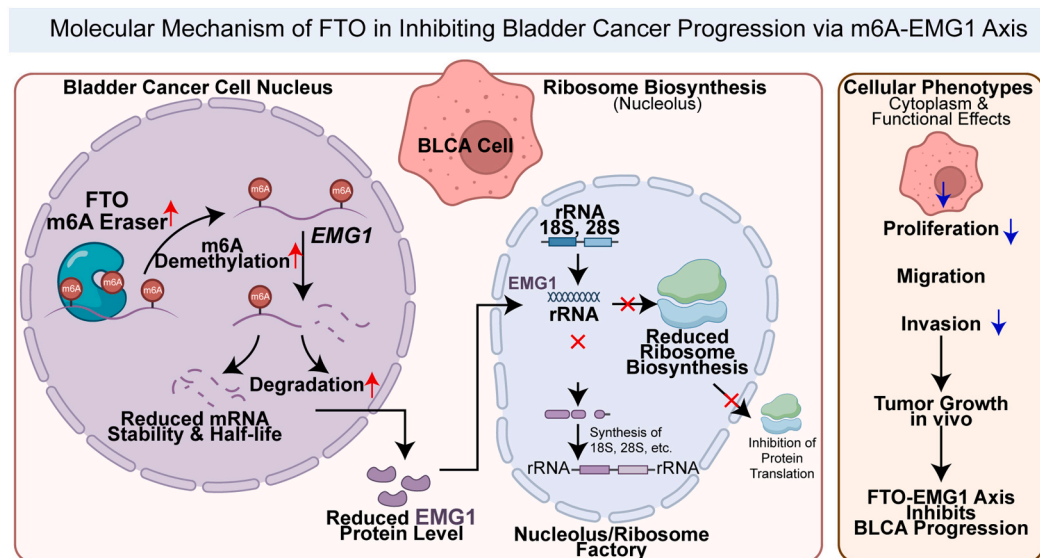


Fig. 10. Schematic mechanism of FTO as a tumor suppressor in bladder cancer via EMG1-mediated inhibition of ribosome biogenesis.

bladder cancer tissues [16,35,36]. This discrepancy may reflect the heterogeneity of bladder cancer, as FTO expression could be stage- or subtype-dependent. Our functional data demonstrated that FTO overexpression inhibited bladder cancer cell proliferation, migration, and invasion, while FTO knockdown had the opposite effect. Moreover, m6A is reported to promote ribosome biogenesis through m6A modification of ribosome biogenesis-associated genes [25], therefore, the role of FTO on ribosome biogenesis within bladder cancer cells was investigated, and it was revealed that FTO overexpression notably repressed nascent rRNA synthesis, whereas FTO knockdown promoted nascent rRNA synthesis, indicating that FTO inhibits rRNA synthesis. These results indicate that FTO is downregulated in bladder cancer, serving as an anti-tumor factor, possibly through a ribosome biogenesis-related manner. Therefore, it is of great significance to study the interaction mechanism between FTO and ribosome biogenesis in bladder cancer.

After confirming FTO regulation of ribosome biogenesis, the related molecular mechanism was further investigated in bladder cancer. Based on bioinformatics analysis, EMG1, RBIS, and DDX3X are key genes involved in ribosome biogenesis in bladder cancer. Since ribosome biogenesis was previously identified as negatively regulated by FTO, and the results of pathway enrichment analysis showed that only EMG1 was enriched in the ribosome biogenesis pathway, EMG1 is identified as the core factor of ribosome biogenesis regulated by FTO. Reportedly, EMG1 exerts a pivotal effect on eukaryotic ribosome biogenesis [37,38]. Moreover, studies have shown that knocking down EMG1 can inhibit the proliferation, migration, invasion, and tumorigenicity of lung adenocarcinoma cells [39]. Taken together, FTO might affect bladder cancer proliferation, and migratory and invasive capabilities by down-regulating EMG1 and ribosome biogenesis. Therefore, the total m6A and EMG1 mRNA levels were detected following FTO overexpression or knockdown, suggesting that FTO overexpression reduced the total m6A and EMG1 mRNA levels and inhibited the m6A level of EMG1. Moreover, FTO also decreased the half-life of EMG1.

Since FTO regulates EMG1 mRNA expression, the dynamic effects of FTO and EMG1 were investigated next. EMG1 is increased within lung cancer and could promote cell proliferation, migration, and invasion, and exert an oncogenic effect on lung cancer [39]. Furthermore, EMG1 is reported to be involved in ribosome biogenesis by regulating EMG1 levels. Furthermore, functional experiments have shown that FTO-mediated EMG1 downregulation inhibited bladder cancer cell proliferation, migratory and invasive capabilities in vitro, and repressed tumor growth in vivo.

In conclusion, the data from this study reveals that FTO is down-regulated in bladder cancer and plays a tumor suppressor role by reducing EMG1 levels and ribosome biogenesis function in bladder cancer in a ribosome biogenesis-dependent manner (Fig. 10). Therefore, FTO may be an important tumor suppressor gene in the development of bladder cancer, and restoring FTO functions might be a promising treatment method in bladder cancer therapy.

CRediT authorship contribution statement

Kun Yao: Writing – review & editing, Writing – original draft, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Long Wang:** Writing – original draft, Project administration, Methodology, Investigation, Formal analysis, Data curation. **Jinrong Wang:** Visualization, Validation, Software, Resources, Methodology, Investigation. **Jianye Liu:** Resources, Methodology, Investigation, Formal analysis, Data curation. **Chao Li:** Software, Methodology, Formal analysis, Data curation. **Weibin Hou:** Visualization, Validation, Methodology, Investigation. **Bingzhi Wang:** Software, Resources, Methodology. **Wei Xiong:** Methodology. **Jing Tan:** Writing – review & editing, Writing – original draft, Supervision, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

Not applicable.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.bbamcr.2026.120147>.

Data availability

The data supporting the findings of this study are available within the article, its supplementary material, and the Figshare repository. The Figshare data can be accessed via the following link <https://doi.org/10.6084/m9.figshare.31553539>.

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